#0A55

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Not Only Agent Screen Saver

PROJECT BRIEF v2.0 | 2026-03-20 | CLASSIFIED: INTERNAL

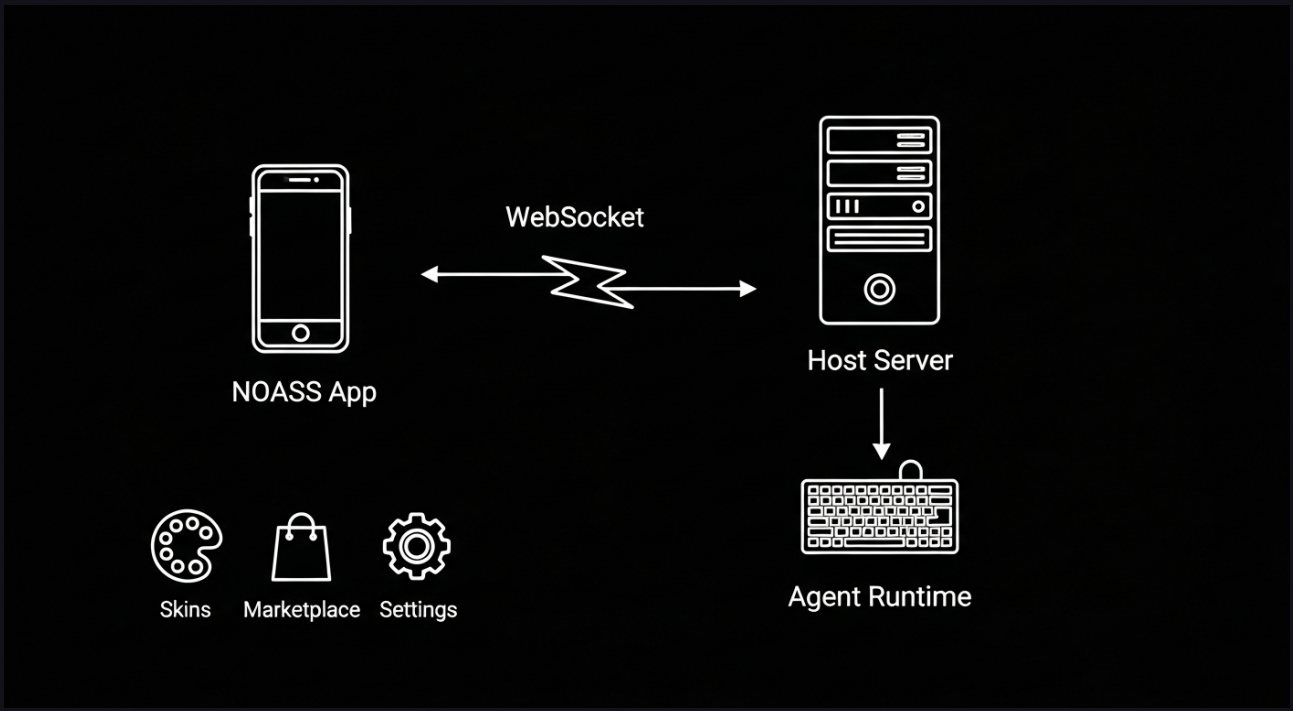
1-bit dithered mobile dashboard for monitoring AI agent sessions. Skins. Physics. Weather. Marketplace. Modules. Floyd-Steinberg everything.

TASKS	STAGES	SKINS	MODULES	RISK
15	6	6	5	HIGH

TECH STACK

Tauri 2	Native shell + WebView for Android/iOS
Vite 6 + TS	Build tooling + type safety
Canvas API	Pure canvas rendering, no framework
Floyd-Steinberg	1-bit dithering on everything
WebSocket	Real-time connection to host server
DeviceMotion	Accelerometer physics + weather

ARCHITECTURE



NOASS is a pure WebSocket client. The server stays on your machine, polling the agent runtime every 2 seconds and broadcasting JSON state. The mobile app connects via Tailscale or local network, renders everything on a single HTML Canvas with Floyd-Steinberg dithering, and sends commands back.

PROTOCOL

Server -> Client (every 2s):

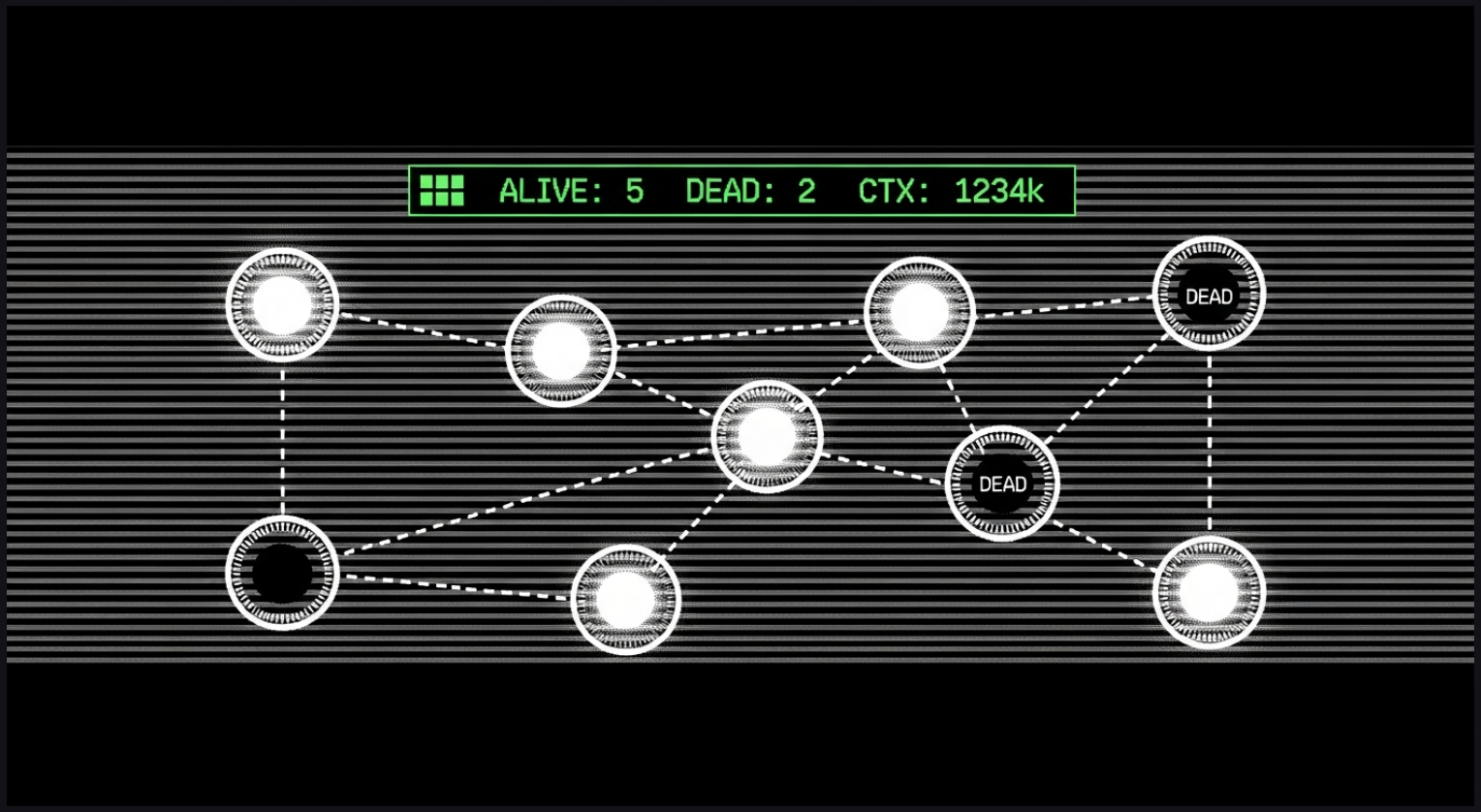
```
{ type: "state", panes: [{name, alive, ctx_k,
  rate_k_per_min, eta_800k_min, ctx_pct, last}],
  log: [...], stats: {...} }
{ type: "readResult", target, content }
{ type: "ack", command, target, success }
```

Client -> Server:

```
{ type: "nudge"|"kill"|"read"|"revive",
  target, message?, lines? }
```

KEY INVARIANTS

- pane.name is the canonical stable identity key
- 30fps target, 3-tier degradation on Android
- Settings uses DOM overlay for reliable keyboard
- Fonts bundled locally (no CDN dependency)
- Modules get ModuleContext, not raw AppState



STAGED PLAN

STAGE 1: CORE	Tauri 2 scaffold + Canvas renderer	T01-T04
STAGE 2: THEMING	Skin system + settings UI	T05-T06
STAGE 3: MARKETPLACE	Skin marketplace + navigation	T07-T08
STAGE 4: BUILD	Android APK + iOS prep + polish	T09-T12
STAGE 5: PHYSICS	Accelerometer + weather effects	T13-T14
STAGE 6: EEOAO	Module system + gamification	T15

TASK BREAKDOWN

T01	Tauri 2 Project Scaffold	**LOW**	**READY**
T02	Port Canvas Renderer	**MED**	**READY**
T03	Port Graph, UI, Controls	**MED**	**READY**
T04	Main App Entry + WS Client	**MED**	**READY**

T05	Default Skins Bundle (6)	LOW	READY
T06	Settings Screen (DOM)	LOW	READY
T07	Marketplace Screen	MED	READY
T08	Navigation + Gestures	MED	READY
T09	Mock Data Mode (Demo)	LOW	READY
T10	Android Build Setup	HIGH	TODO
T11	iOS Project Prep	MED	TODO
T12	Polish & Ship	MED	TODO
T13	Accelerometer Physics	MED	TODO
T14	Weather Effects	MED	TODO
T15	EEOAO Module System	HIGH	TODO

SKIN SYSTEM

Every visual parameter is controlled by a Skin manifest: accent color, node grays, scanline opacity/gap, vignette strength, CRT curvature, pulse frequency, font family, connection style, weather type, and device gravity strength.

BUNDLED SKINS:

Matrix	#00ff41	Classic green phosphor terminal
Amber CRT	#ffb000	Warm amber phosphor. 1983 IBM.
Ice	#00d4ff	Cold blue. Server room at 3AM.
Blood Moon	#ff2222	Red alert. Something is on fire.
Phantom	#bb86fc	Pale violet. Ghost in the machine.
Solar Flare	#ff6600	Orange nova. High energy.

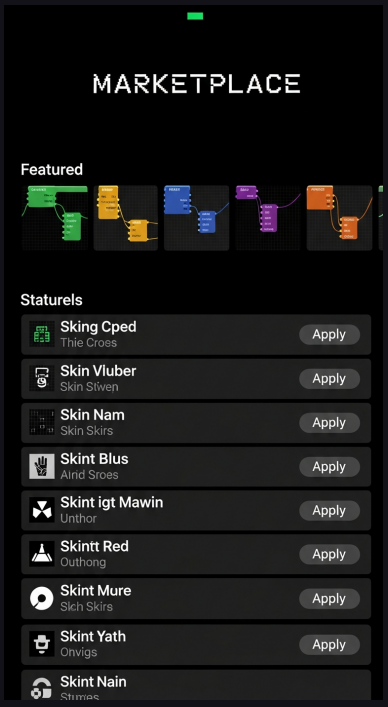
PHYSICS & WEATHER

Accelerometer physics: tilt the phone, nodes slide with gravity. Shake to scatter. Normalized screen-space gravity with smoothing. 2-second shake cooldown. iOS permission via settings toggle.

WEATHER TYPES:

Rain	Vertical white streaks, tilt-reactive
Snow	Slow drift, accumulates at bottom
Static	TV noise, density via skin
Matrix	Falling green chars, classic
Sparks	Float upward from alive nodes

MARKETPLACE



Browse, preview, and apply skins. Featured row + grid layout. Tap to live-preview (entire UI changes to show skin in action). Bundled catalog for MVP, remote catalog API later.

EEOAO: MODULE SYSTEM

Everything Everywhere All at Once. Each project becomes a module inside #0A55. Modules get isolated ModuleContext (read-only host state, namespaced storage). Lifecycle: destroy -> init, no leaked side effects.

PLANNED MODULES:

Orchestra	NeoOrchestra/Strudel live coding audio
Voice	ReVisper speech-to-text
Ghost Hunt	Dithered co-op game viewer
HackRF	Spectrum dashboard
Crawler	Web crawler status

ACHIEVEMENTS (unlock hidden modules):

First Blood	Kill an agent from dashboard
Night Owl	Use app after midnight
Context Lord	Monitor 1M+ total context
Skin Collector	Apply 3 different skins

ARCHITECT REVIEW (GPT 5.4)

ROUND 1: Stages 1-4		
5 BLOCK	2 DEGRADE	RESOLVED

ROUND 2: Stages 5-6		
6 BLOCK	2 DEGRADE	RESOLVED

ROUND 1 - BLOCKING RESOLVED:

[INTERFACE] read flow missing response contract
FIX: readResult/ack already in server.js, added to types

[COUPLING] thin shell vs mobile-native
FIX: All web APIs except back button (Tauri)

[INVARIANT] node identity underspecified
FIX: pane.name canonical, reconciliation explicit

[RESOURCE] no performance budget
FIX: 30fps, 3-tier degradation

[INTERFACE] settings canvas text input
FIX: DOM overlay instead

ROUND 2 - BLOCKING RESOLVED:

[TEMPORAL] iOS motion permission
FIX: Settings toggle calls requestPermission on tap

[INVARIANT] gravity vector unnormalized
FIX: Screen-space mapping, clamp [-1,1], smoothing

[RESOURCE] degradation only covers particles
FIX: Extended: weather->physics->dither->vignette

[INTERFACE] module IDs unconstrained
FIX: Regex /^[a-z][a-z0-9-]{1,30}$/, reject reserved

[TEMPORAL] module lifecycle unordered
FIX: destroy() before init(), onDestroy() hooks

[INTERFACE] shared AppState unconstrained
FIX: ModuleContext: read-only host, namespaced store

SURVIVABILITY

15 tasks across 6 stages. All 11 blocking critiques resolved. Highest risk: Task 10 (Android SDK on WSL2) and Task 15 (first real module port tests the isolation contract). Physics/weather need real-device accelerometer testing. Plan survives contact with the codebase.